

Part One Concept Reference

Central concepts and notation for Chapters 1–9

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This reference collects selected concepts from Chapters 1–9 and the notation those chapters use. Its six concept tables work at two levels: a broad **concept group** gathers ideas that belong together, and within a group a selected **term** is listed. Each row displays four things: the concept group, the term with its common alternatives, its meaning or the distinction that matters most, and the main explanation, the place in the book where that meaning is developed. A closing table lists the notation of Chapters 1–9 with each symbol’s meaning, main use, and alternatives. The reference is meant to be consulted while reading rather than read through: a term’s first use, its worked example, and the reasoning around it stay in the chapter that teaches it.

A few conventions govern the rows. Names that pick out the same concept share one entry, so *mean* and *arithmetic mean* appear together, and *outcome*, *response*, and *dependent variable* are explained as one analytical role. A more specific role, such as treatment, exposure, or confounder, carries additional design or substantive meaning that the study itself must supply. Related concepts with different meanings, such as mean and median or SD and SE, keep separate entries. *Main explanation* names the principal teaching home rather than a first mention, so a brief early introduction and a later full account can coexist: Chapter 1 introduces cases and samples through its studies, while Chapter 2 develops their use in reading records

and calculations. A few rows cite *Sup. A* and *Sup. B*, the two supplementary units *How to Read a Quantitative Study as an Argument* and *Extending the Argument-Based Approach to Quantitative Studies*; neither unit is yet available.

1 Questions, Study Design, and Data

Table 1

Questions, study design, and data.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Questions and arguments	Question-answer pair	What a study asks and the answer it eventually offers for that question. Each separately owed substantive answer has its own argument.	Ch. 1, Sec. 1.3
Questions and arguments	Traceable information	Specific study content through which an argument component's function can be identified at a documentary location. Identifying that contribution precedes judging its adequacy.	Ch. 1, Sec. 6.2
Questions and arguments	Estimand	The target quantity sought for the named units, conditions, outcome, and scope. It can concern a fixed dataset, a population, a comparison, or prediction performance.	Ch. 1, Sec. 1.3
Questions and arguments	Evidence	Observations and contextual records contributing to an answer, including information about how they were produced.	Ch. 1, Sec. 6.1
Questions and arguments	Justification	Reasons and conditions invoked to support a comparison, analytical move, or interpretation.	Ch. 1, Sec. 6.1
Questions and arguments	Inference	Reasoning and analysis connecting evidence and justification to a conclusion.	Ch. 1, Sec. 6.1
Questions and arguments	Bounded claim	The reported conclusion with its uncertainty, conditions, and scope: in the statistical layer, the conclusion about the estimand; in the substantive layer, the question-facing answer with its qualifications.	Ch. 1, Sec. 6.1
Data and variables	Unit	The entity selected, assigned, measured, or analyzed in a particular operation. The units of assignment, measurement, and analysis can differ, and all are distinct from measurement units such as points or minutes.	Ch. 1, Sec. 2.1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Data and variables	Case	The entity or measurement occasion described in a dataset. Depending on the study, a case may be a student, a class, or one measurement occasion for a student.	Ch. 2, Sec. 1
Data and variables	Variable	A characteristic that can take different values across cases or occasions. A dataset records values of one or more variables for its cases.	Ch. 2, Sec. 1
Data and variables	Quantitative variable (numerical variable)	A variable expressing an amount, including continuous measurements and discrete counts or scores. Its units and scale determine what numerical differences mean.	Ch. 2, Sec. 1
Data and variables	Categorical variable	A variable whose values identify categories. Nominal categories have no intrinsic order; ordinal categories have an order; a binary variable has two categories.	Ch. 2, Sec. 1
Variable roles	Outcome variable (response variable; dependent variable)	The variable an analysis seeks to describe, compare, or predict. These alternative names share that role in many analyses; a variable's role depends on the question.	Ch. 1, Sec. 2.1
Variable roles	Predictor variable (explanatory variable; independent variable)	A variable used to represent an outcome's relationship with other information. These analytical-role labels alone establish neither a causal effect nor statistical independence.	Ch. 1, Sec. 2.1
Study scope	Population	The collection of units a question concerns. It may be the participating students themselves or a wider collection to which the findings might extend.	Ch. 2, Sec. 1
Study scope	Sample	A selected set of units from a larger collection. Calling a set a sample does not by itself explain how it was selected or which population it represents.	Ch. 2, Sec. 1
Study scope	Analysis sample	The units whose data contribute to a particular analysis. Eligibility, missing measurements, and exclusions can make this set and its denominator differ across analyses.	Ch. 2, Sec. 1
Study design	Sampling (recruitment; selection into a study)	How units come to enter a study, by probability selection, volunteering, or availability. It is contrasted with assignment, which concerns the conditions allocated to participating units.	Ch. 1, Sec. 3.2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Study design	Assignment (allocation)	The allocation of study conditions to participating units, by any mechanism. Actual treatment received may differ from the assigned condition.	Ch. 1, Sec. 3.2
Study design	Random sampling (probability sampling)	Selecting units from a defined population or sampling frame through a probability mechanism that gives every eligible unit a known, nonzero chance of selection, not necessarily the same chance. “Random selection” alone can describe other chance-based decisions.	Ch. 1, Sec. 3.2
Study design	Random assignment (random allocation)	Allocating study conditions to participating units through a specified probability mechanism. The assigned units may be individuals or groups; realized groups need not be identical.	Ch. 1, Sec. 3.2
Study design	Causal effect (counterfactual contrast)	The difference between the outcomes that would occur for the same units or target population under two specified conditions, only one of which is observed for each unit. The contrast may be zero; interpreting a comparison causally needs design and identification support.	Ch. 1, Sec. 3.1
Study design	Sampling design (sampling scheme)	The specified procedure for selecting a sample, including selection stages, grouping, and, for probability sampling, the selection probabilities.	Ch. 2, Sec. 1.5
Study design	Simple random sampling	Selection in which every possible sample of the specified size has the same probability of being chosen.	Ch. 2, Sec. 1.5
Study design	Confounding	Mixing an exposure’s effect with effects of other factors that influence both exposure and outcome. Such a factor is a confounder, a specific causal role.	Ch. 1, Sec. 3.1
Study design	External validity (generalizability)	Support for extending findings to other people, settings, or conditions. The intended extension determines which similarities and differences matter.	Ch. 1, Sec. 3.2
Argument-based reading	Argument component	One of the four functions, evidence, justification, inference, and bounded claim, that a piece of study information performs in the statistical or the substantive layer. A reader identifies the function before judging its adequacy.	Sup. A, Sec. 2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Argument-based reading	Documentary location	The place in a report or its related materials, such as Methods, a protocol, a codebook, or an analytical supplement, where the traceable information for a component is found.	Sup. A, Sec. 4
Argument-based reading	Evaluation question	One of the six questions on alignment, support, and translation used to judge the quality of a question-level argument after it has been read. Each receives a reasoned judgment, not a score.	Sup. A, Sec. 5
Argument-based reading	Statistical strand	An estimand-specific statistical chain inside one argument. Two strands arise only when sibling estimands are jointly required for one integrated substantive answer; each has its own statistical inference and conclusion.	Sup. B, Sec. 1
Argument-based reading	Coordinated questions	Several question-answer pairs in one study whose support depends on one another. Each answer is read on its own, and their coherence across questions is judged separately.	Sup. B, Sec. 3

2 Descriptions, Distributions, and Sampling Variation

Table 2

Descriptions, distributions, and sampling variation.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Observed distributions	Empirical distribution (observed distribution; data distribution)	How the observed values are distributed across values or intervals, including their frequencies, center, spread, and shape. A probability model is a separate description of possible values.	Ch. 2, Sec. 2
Center and spread	Mean (arithmetic mean)	The sum of the numerical values divided by their count. Every included value contributes, so unusual values can shift the mean.	Ch. 2, Sec. 2
Center and spread	Median	The middle value after ordering the data, or the average of the two middle values when the count is even. It describes the center by position.	Ch. 2, Sec. 2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Center and spread	Quartile	A cut point dividing ordered data into quarters. The first and third quartiles mark approximately the lower and upper edges of the middle half; calculation conventions can differ.	Ch. 2, Sec. 2
Center and spread	Interquartile range (IQR)	The third quartile minus the first quartile: the width of the middle half of the ordered data, expressed in the variable's original units.	Ch. 2, Sec. 2
Center and spread	Variance	A measure of spread based on squared deviations from the mean. The usual sample variance divides their sum by $n - 1$ and is expressed in squared units.	Ch. 2, Sec. 2
Center and spread	Standard deviation (SD)	The square root of the variance, expressing spread around the mean in the variable's original units. Applied to individual observations, it describes variation among their values; applied to a statistic's sampling distribution, it is the SE.	Ch. 2, Sec. 2
Group descriptions	Grand mean	The mean across all included cases. It equals the average of the group means weighted by their group counts, so larger groups contribute more.	Ch. 2, Sec. 2
Variation and constraints	Degrees of freedom (df)	A count reflecting freely varying information after constraints or fitted quantities are taken into account. The n deviations from their own mean sum to zero, leaving $n - 1$ free.	Ch. 2, Sec. 2
Probability models	Standardized value (z score)	A value minus a specified mean, divided by a specified standard deviation. It expresses distance in SD units; standardizing alone preserves the distribution's shape.	Ch. 2, Sec. 2
Probability models	Normal distribution (Gaussian distribution)	A symmetric, bell-shaped probability model described by a mean and standard deviation. The standard Normal has mean zero and SD one; observed data need not follow this model.	Ch. 2, Sec. 2
Relationships	Conditional proportion	The proportion in one category among cases satisfying a stated condition. Its denominator is the count in that conditioning group, such as students in one education-intention category.	Ch. 2, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Relationships	Covariance	A summary based on products of paired deviations from the two means. Its sign shows the direction of joint variation; its magnitude depends on both variables' units and spreads.	Ch. 2, Sec. 3
Relationships	Pearson's correlation (Pearson's r ; product-moment correlation)	Covariance divided by the product of the two SDs. For variables with nonzero spread, it describes direction and strength of linear association on a scale from -1 to 1.	Ch. 2, Sec. 3
Sampling variation	Sampling distribution	The distribution of a statistic, such as a sample mean, across repeated samples generated by a stated sampling procedure or model. Repeated sample summaries in a simulation approximate this distribution.	Ch. 2, Sec. 4
Sampling variation	Sampling error	The part of an estimate's departure from the population quantity that arises from observing a sample rather than the whole population. It varies across repetitions, and its spread is the SE; coverage, nonresponse, and measurement error are separate.	Ch. 2, Sec. 4.3
Sampling variation	Standard error (SE)	The standard deviation of a statistic's sampling distribution under a stated sampling procedure or model. It describes spread across repeated sample summaries, distinct from individual-value SD and one sample's difference from the population quantity.	Ch. 2, Sec. 4
Data structure	Data matrix	A rectangular arrangement with cases in rows and variables in columns; x_{ij} denotes case i on variable j .	Ch. 2, Sec. 1
Variables	Binary variable (0/1 coding)	A binary variable has two possible categories; a 0/1 coding marks membership or occurrence. Its sample mean is the proportion coded 1.	Ch. 2, Sec. 2
Transformations	Linear transformation	$Y = a + bX$ changes the center to $a + bE(X)$ and the SD to $ b SD(X)$; a negative b also reverses order.	Ch. 2, Sec. 2
Distribution models	Bernoulli; binomial; categorical; Poisson; lognormal; logistic	Respectively a binary outcome, a count of successes in fixed independent equal-probability trials, one category, an event count under a specified rate model, a positive variable with Normal log, and a continuous symmetric distribution used here as a preview. Logistic regression later models Bernoulli outcomes.	Ch. 2, Sec. 2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Probability	Density; cumulative distribution function; quantile	Density determines probability by area for a continuous variable; the CDF gives $P(X \leq x)$; a quantile is a value reaching the stated cumulative probability.	Ch. 2, Sec. 2
Relationships	Rank correlation; point-biserial correlation	Spearman uses ranks; Kendall summarizes concordance; point-biserial is Pearson correlation with one binary variable coded 0/1. Each addresses a specified relationship.	Ch. 2, Sec. 3
Relationships	Partial correlation	Correlation between two variables after their linear relationships with stated controls are removed by regression; it describes residual association and does not establish causal adjustment.	Ch. 2, Sec. 3
Data summaries	Covariance matrix	A square table of variances on the diagonal and covariances off the diagonal for a stated set of variables and observations.	Ch. 5, Sec. 5
Data summaries	Correlation matrix	A square table of pairwise correlation coefficients for a stated set of variables and observations; its diagonal is 1 for variables with positive variance.	Ch. 5, Sec. 5

3 Estimation, Uncertainty, and Statistical Testing

Table 3

Estimation, uncertainty, and statistical testing.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Estimation	Estimate	A numerical approximation to an estimand, calculated from the available data. A point estimate expresses the approximation as one value.	Ch. 3, Sec. 1
Estimation	Estimator	The rule used to calculate an estimate from data. Its bias and precision concern how the rule behaves across the repetitions specified by the study's procedure or model.	Ch. 3, Sec. 1
Estimation	Bias (of an estimator)	The difference between an estimator's expected value over the specified repetitions and the estimand. An unbiased estimator has bias zero; precision describes its spread, a separate quality.	Ch. 3, Sec. 1

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Estimation	Parameter	A numerical characteristic of a population or probability model, such as a population mean. A parameter may be the estimand; the question determines which quantity is sought.	Ch. 2, Sec. 4
Estimation	Statistic	A quantity calculated from sample data, such as a sample mean or proportion. The same statistic can summarize the sample or serve as an estimate for a wider target.	Ch. 2, Sec. 4
Uncertainty	Confidence interval (CI)	An interval expressing uncertainty about an estimand. Under its conditions, the procedure covers that estimand at its stated rate across repetitions; the rate describes the procedure.	Ch. 3, Sec. 3
Uncertainty	Critical value	The cut-point of a reference distribution, such as the Normal or t, selected by a confidence or significance level. It multiplies the SE in an interval and marks the rejection region of a test.	Ch. 3, Sec. 3
Uncertainty	Coverage	The proportion of repetitions in which an interval procedure contains the estimand under its conditions. The confidence level is the procedure's target coverage, not a probability for one realized interval.	Ch. 3, Sec. 3
Statistical testing	Null hypothesis	The statistical proposition used as the reference for a test, such as equality of specified population means. The alternative hypothesis states the departure being considered.	Ch. 3, Sec. 4
Statistical testing	Test statistic	A quantity calculated from data to assess departure from a null hypothesis. The test compares it with a reference distribution justified under that null and stated conditions.	Ch. 3, Sec. 4
Statistical testing	p-value	Under a specified null hypothesis and conditions, the probability of a test result at least as extreme as the observed one, using the test's stated tail rule.	Ch. 3, Sec. 4
Statistical testing	Significance level (alpha)	The threshold used for rejecting a null hypothesis, chosen to control the probability of rejecting a true null under the test's conditions. This error is called a Type I error.	Ch. 3, Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Statistical testing	Expected count	The count a cell of a two-way table would have under a stated null pattern such as independence, given the observed margins: row total times column total, divided by the overall total.	Ch. 3, Sec. 4
Statistical testing	Pearson's chi-square statistic (X^2)	The sum over cells of squared differences between observed and expected counts, each divided by its expected count, compared with a chi-square reference with $(R - 1)(C - 1)$ degrees of freedom under the stated model.	Ch. 3, Sec. 4
Interpreting tests	Power (statistical power)	The probability of rejecting the null under a specified alternative, design, and test. Failure to reject under that alternative is a Type II error; power is one minus its probability.	Ch. 3, Sec. 5
Several comparisons	Familywise error rate (FWER)	The probability of at least one Type I error among a defined family of tests. Procedures such as Bonferroni allocate the error allowance across that family.	Ch. 3, Sec. 5
Tests for means	One-sample z test; one-sample t test	Both compare a sample mean with a stated null mean using its SE; z uses a known process SD under its model, whereas t estimates that SD and uses the appropriate t reference. Neither choice follows an automatic sample-size cutoff.	Ch. 3, Sec. 4
Tests for means	Independent-samples t test; Welch test; pooled t test	Independent-group tests compare means. Welch allows unequal variances with an estimated reference df; the pooled procedure adds the equal-variance condition.	Ch. 3, Sec. 4
Tests for means	Paired-samples t test	A one-sample t test of within-pair differences against a specified mean difference, with independent pairs under the stated model.	Ch. 3, Sec. 4

4 Group Comparisons and ANOVA

Table 4

Group comparisons and analysis of variance.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Comparison design	Factor (grouping factor)	In ANOVA, a categorical variable defining groups or conditions; its categories are levels. This meaning differs from a latent factor used to model shared indicator variation.	Ch. 4, Sec. 1
ANOVA	Analysis of variance (ANOVA)	A method for studying mean differences through sources of variation. One-way and two-way ANOVA use one and two grouping factors, respectively; either factor can have several levels.	Ch. 4, Sec. 2
Model quantities	Fitted value	The value a fitted model assigns to an observed case. In a one-way group-mean model, it is that case's group mean.	Ch. 4, Sec. 2
Model quantities	Residual	An observed outcome minus its fitted value. A residual is calculated from the fitted model; an error is the corresponding unknown departure in the population model.	Ch. 4, Sec. 2
ANOVA quantities	Mean square (MS)	A sum of squares divided by its degrees of freedom. In classical ANOVA, the within-group mean square estimates the common error variance under the model's conditions.	Ch. 4, Sec. 2
ANOVA quantities	F statistic (F ratio)	In classical ANOVA, a ratio of the mean square for a tested source to the appropriate error mean square. Its null F reference depends on the model and both degrees of freedom.	Ch. 4, Sec. 2
ANOVA quantities	Omnibus test	A test of one joint null hypothesis about a whole pattern, such as equality of all group means. Rejecting it does not identify which comparisons differ or how large they are.	Ch. 4, Sec. 2
Magnitude	Standardized mean difference	A difference in means divided by a specified SD. The chosen denominator determines the standardized quantity; the raw difference retains the outcome's original units.	Ch. 4, Sec. 2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Magnitude	Eta squared	The proportion of total observed variation attributed to a specified source in an ANOVA decomposition. In one-way ANOVA, it is between-group divided by total sum of squares.	Ch. 4, Sec. 2
Specific comparisons	Contrast (linear contrast)	A weighted combination of means whose weights sum to zero, expressing a specific comparison. The means, weights, and target population determine the quantity compared.	Ch. 4, Sec. 3
Specific comparisons	Simultaneous confidence intervals	A set of intervals calibrated so that, under their conditions, all of them cover their estimands together at the stated familywise rate; Bonferroni, Tukey, and Dunnett are different calibrations for different families.	Ch. 4, Sec. 3
Factorial comparisons	Cell mean and marginal mean	In a two-factor layout, a cell mean is the mean of the cases sharing one level of each factor; a marginal mean averages the cell means across the levels of the other factor, with stated weights.	Ch. 4, Sec. 4
Factorial comparisons	Main effect	A comparison across one factor's levels averaged over the other factor's levels, using stated weights. It remains an average when the conditional pattern differs across levels.	Ch. 4, Sec. 4
Factorial comparisons	Interaction	A pattern in which a comparison across one factor's levels changes with the other factor's level. A difference of differences expresses one such comparison.	Ch. 4, Sec. 4
Factorial comparisons	Simple effect	A comparison across one factor's levels at a specified level of the other factor. It describes a conditional comparison within the factorial design.	Ch. 4, Sec. 4
Adjusted comparisons	ANCOVA (analysis of covariance)	A linear model comparing groups while including quantitative covariates; its adjusted comparison is defined at stated covariate values or over a stated covariate distribution.	Ch. 6, Sec. 4

5 Statistical Computing

Table 5
Statistical computing.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
R basics	Object (R object)	A named value, vector, table, model result, or other item stored in an R session.	Ch. 5, Sec. 1
R basics	Vector	Assignment with <code><-</code> names an object; the Environment pane lists the current objects. A sequence of values of one basic type: numeric, character, or logical. <code>NA</code> marks an unavailable value; it is not zero. A data frame is a table whose columns are vectors of equal length, one row per case.	Ch. 5, Sec. 1
R basics	Function and argument (R function)	A function performs an operation on supplied inputs; an argument names an input or an option, such as <code>na.rm = TRUE</code> . An R argument is distinct from a study argument in the argument-based approach.	Ch. 5, Sec. 1
R basics	Package (library)	An installed collection of functions and data, such as stats or graphics, which are supplied with R. Installing an additional package makes it available on the computer; <code>library()</code> attaches it for the session; <code>package::function()</code> names one exported function explicitly.	Ch. 5, Sec. 1
Reproducible work	Script and Quarto source	A saved <code>.R</code> script records commands so an analysis can be rerun; a <code>.qmd</code> source connects prose, mathematics, executable code, tables, figures, and references, and renders to PDF, HTML, or Word.	Ch. 5, Sec. 1
Reproducible work	Model formula (R formula)	Notation such as <code>score ~ group</code> naming an outcome and the variables that explain it for a fitting function. It does not state whether observations are independent or paired; that comes from the design.	Ch. 6, Sec. 1
Simulation	Simulation and seed (Monte Carlo simulation)	Generating data from a declared process to see what a statistic or procedure does across repetitions. A seed makes the random draws repeatable; it does not make the process true for an actual study.	Ch. 5, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Simulation	Monte Carlo standard error	The numerical uncertainty of a simulated probability or mean that comes from using finitely many repetitions; more repetitions shrink it. It is distinct from a study's SE, which depends on the study's sample size.	Ch. 5, Sec. 4
Numerical fitting	Objective function	A numerical criterion evaluated at candidate parameter values for fixed data; fitting seeks a stated minimum or maximum.	Ch. 6, Sec. 6
Numerical fitting	Likelihood and log-likelihood	The probability or density of the observed data under a declared model, viewed as a function of its parameters; the log-likelihood is its logarithm.	Ch. 6, Sec. 6; fuller Ch. 11, Sec. 3
Numerical fitting	Maximum likelihood estimation (MLE)	Choosing parameter values that maximize the likelihood, equivalently the log-likelihood, over the stated parameter space.	Ch. 6, Sec. 6; fuller Ch. 11, Sec. 3

6 Regression

Table 6
Regression.

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
The fitted line	Simple linear regression (regression line)	A straight-line summary of how one outcome differs across one predictor: fitted value equals intercept plus slope times the predictor. Its descriptive use concerns the records fitted; a model-based use needs a declared process.	Ch. 7, Sec. 2
The fitted line	Intercept	The fitted outcome at predictor value zero. It places the line vertically; when zero is outside the observed support, it is a mathematical anchor rather than a supported prediction. Centering the predictor moves it to a supported reference point.	Ch. 7, Sec. 2

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
The fitted line	Slope (regression coefficient)	The difference in fitted outcome per one-unit difference in the predictor, in outcome units per predictor unit. It compares units that differ in the predictor; it is not a change within one unit unless a design supports that reading.	Ch. 7, Sec. 2
Least squares	Ordinary least squares (OLS)	The rule that chooses the intercept and slope minimizing the residual sum of squares. With an intercept, the fitted line passes through the mean point and the residuals sum to zero; the slope equals covariance divided by predictor variance.	Ch. 7, Sec. 2
Least squares	Residual sum of squares (RSS)	The sum of squared residuals from a fitted line, in squared outcome units. Total squared variation about the mean splits into a regression part and RSS when the model has an intercept.	Ch. 7, Sec. 2
Fit	Residual standard deviation (residual standard error)	The square root of RSS divided by its residual degrees of freedom, in outcome units; with an intercept and one predictor the degrees of freedom are $n - 2$. Under the process model it estimates the error SD.	Ch. 7, Sec. 2
Fit	R-squared (coefficient of determination)	One minus RSS divided by total squared variation: the proportion of centered squared outcome variation accounted for by the fit in these records. In simple regression with an intercept it equals the squared correlation.	Ch. 7, Sec. 2
Prediction	Confidence interval for a mean response (mean-response interval)	An interval for the process mean outcome at a stated predictor value, using the SE of the fitted value. It concerns an average, narrows near the predictor mean, and does not cover individual outcomes.	Ch. 7, Sec. 5
Prediction	Prediction interval (new-case interval)	An interval for one new outcome at a stated predictor value from the same process. Its variance adds the error variance to the fitted-mean variance, so it is wider than the mean-response interval and keeps a nonzero floor as the sample grows.	Ch. 7, Sec. 5
Checks	Leverage	A measure of how unusual a record's predictor value is relative to the others; it grows with distance from the predictor mean and appears in the SE of a fitted value and in influence measures.	Ch. 7, Sec. 6

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Checks	Cook's distance (influence)	A summary of how much the fitted values change when one record is omitted, combining its residual and its leverage. It prompts inspection of provenance and a disclosed sensitivity, not automatic deletion.	Ch. 7, Sec. 6
Checks	Heteroskedasticity-consistent covariance (HC3; sandwich estimator)	A coefficient covariance estimate that allows unequal error variance across predictor values under independence. It changes SEs and intervals, not the fitted coefficients, and does not repair curvature, dependence, measurement error, or confounding.	Ch. 7, Sec. 6
Checks	Extrapolation (predictor support)	Using a fitted line at predictor values outside the observed range or where observations are sparse. Software returns a number there; the data do not support it.	Ch. 7, Sec. 6
Several predictors	Multiple regression (additive model)	A fitted equation in which the outcome's mean is the sum of an intercept and one term per predictor. Each coefficient compares fitted means at the same values of the other predictors; the additive form assumes each comparison is the same across those values.	Ch. 8, Sec. 2
Several predictors	Partial slope (conditional coefficient; adjusted coefficient)	The difference in fitted outcome per one-unit difference in one predictor, holding the model's other predictors fixed. Its meaning depends on the conditioning set named with it; it is not an intervention effect or a claim about unmeasured differences.	Ch. 8, Sec. 2
Several predictors	Indicator variable (dummy variable) and reference category	A 0/1 variable marking membership in one category of a factor; a factor with J levels needs $J - 1$ indicators, and the omitted level is the reference. The intercept refers to the reference category; each indicator coefficient is a difference from it at the same other predictors. Releveling changes coordinates, not fitted values.	Ch. 8, Sec. 2; fuller Ch. 9, Sec. 1
Several predictors	Partial regression (residualization; added-variable form)	Regressing both the outcome and one predictor on the other predictors, then relating their residuals. The residual slope equals the partial slope; the residual correlation is the partial correlation, unit-free.	Ch. 8, Sec. 3

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Uncertainty	Coefficient covariance matrix	The estimated variances and covariances of the fitted coefficients (<code>vcov()</code> in R). Any linear combination of coefficients, such as a fitted mean or a contrast, has variance $c'\widehat{V}c$; adding SEs or ignoring covariances is a different, incorrect calculation.	Ch. 8, Sec. 5
Checks	Variance inflation factor (VIF)	One divided by one minus the R^2 from regressing a predictor on the other predictors. Its square root multiplies that coefficient's SE relative to an orthogonal design, holding residual variance fixed; it describes precision, not bias, and is not a deletion rule.	Ch. 8, Sec. 4
Checks	Joint support (covariate hull)	Whether a combination of predictor values lies within the region where observations actually occur, not merely within each variable's separate range. A fitted value outside joint support is extrapolation even if each coordinate is observed somewhere.	Ch. 8, Sec. 4
Fitted quantities	Standardization over a named distribution (model-averaged fitted value)	Averaging the model's fitted values after setting one predictor to a chosen value for every profile in a declared set with fixed weights. Its uncertainty uses the coefficient covariance; it is not an observed group mean, a causal effect, or held-out performance.	Ch. 8, Sec. 5
Prediction	Learning rule (learning procedure)	A fixed recipe that turns training records into a prediction equation, such as fitting a declared OLS formula. Prediction performance concerns the rule at stated training sizes, not one final fitted equation.	Ch. 8, Sec. 6
Prediction	Cross-validation (held-out validation)	Dividing records into folds, fitting a rule on the other folds, predicting the held-out fold, and pooling the errors so that no outcome contributes to its own prediction. Resplits describe sensitivity to the partition, not new independent studies.	Ch. 8, Sec. 6; fuller Ch. 13, Sec. 2
Prediction	Mean absolute error (MAE)	The average absolute difference between held-out outcomes and their predictions, in outcome units. It is an evaluation loss chosen separately from the squared-error criterion that OLS minimizes.	Ch. 8, Sec. 6

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Coordinated questions	Cross-question coherence	The additional evaluation question for coordinated answers: does the synthesis preserve the distinct questions, estimands, methods, uncertainty, and limits of the separate answers rather than merging them into one omnibus claim?	Ch. 8, Sec. 7
Model comparison	Nested-model F test (extra sum of squares; block test)	Compares a reduced model with a fuller one fitted to the same records and outcome: the RSS reduction per added coefficient, divided by the fuller model's residual mean square, referred to F with the number of restrictions and the fuller model's residual df. It tests the added block jointly, not each coefficient separately.	Ch. 9, Sec. 3
Model comparison	Information criteria (AIC; BIC)	Penalized in-sample fit: $-2 \log \widehat{L}$ plus $2k$ (AIC) or $k \log n$ (BIC), with k counting the coefficients and the residual variance. Smaller is preferred among models fitted to the same records and outcome scale; neither is held-out error.	Ch. 9, Sec. 3
Model form	Polynomial term (quadratic term; curvature)	A power of a predictor added as a further column so that the fitted mean can curve. The predictor's comparison then depends on its value; centering before squaring reduces collinearity without changing the fitted curve.	Ch. 9, Sec. 4
Model form	Log-transformed outcome (geometric-mean scale)	Fitting the logarithm of the outcome, so that a coefficient is an additive difference in expected log outcome and $100(e^\beta - 1)$ percent is a multiplicative difference in geometric means. A changed outcome scale is a changed question, and its criteria are not comparable with the original scale's.	Ch. 9, Sec. 4
Model form	Standardized coefficient (beta weight)	A partial slope rescaled by the predictor and outcome SDs, in outcome-SD units per predictor SD, with the other predictors held fixed. It differs from standardization over named profiles and does not measure causal importance.	Ch. 9, Sec. 4

Concept group	Term and common alternatives	Meaning or essential distinction	Main explanation
Uncertainty	Bootstrap (case resampling; percentile interval)	Refitting the model to many resamples of the records drawn with replacement and using the spread of the refitted estimates as an SE, or their middle 95% as a percentile interval. It treats the resampled units as independent and exchangeable; it does not repair dependence, thin support, measurement error, or an observational design.	Ch. 9, Sec. 6
Uncertainty	Cluster-robust standard error (school-clustered covariance)	A coefficient covariance that sums residual contributions within clusters, such as schools, before combining them, allowing dependence within clusters under an approximation that treats clusters as independent. It changes uncertainty, not the coefficient, and does not repair missing outcomes or weighting.	Ch. 9, Sec. 7

7 Notation for Chapters 1–9

The table below lists the notation Chapters 1–9 use, with each chapter explaining its symbols where they are first used.

Table 7

Notation for Chapters 1–9.

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Data	n, n_1, n_0, n_j, N	Number of cases in the analysis sample; group sizes; a population or record count	Ch. 2	n_A, n_B for named groups in Chapters 5–6
Data	$i, j, k; x_{ij}$	Locally declared indices; a data-matrix entry	Ch. 2	In the data matrix, i is case and j variable. ANOVA declares factor/group/replicate indices beside its own model.
Data	y_i, x_i	Observed outcome and predictor values of case i ; lowercase marks observed values	Ch. 2	y_{ij} for case i in group j

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Data	Y_i, Y_{ij}	The outcome as a random variable before observation; uppercase marks random quantities	Ch. 3	Use only where the distinction from observed values matters
Summaries	$\bar{y}, \bar{x}, \bar{y}_j, \bar{d}$	Observed means; a group mean; the mean of paired differences (write <code>\bar{y}</code> , no braces)	Ch. 2	Use lowercase for observed group means
Summaries	$\bar{Y}$	The sample mean as an estimator in variance and SE derivations	Ch. 3	Reserved for the random quantity
Summaries	$s, s^2, s_1, s_0, s_p, s_d$	Sample SD and variance; group SDs; pooled SD; SD of paired differences	Ch. 2	SD in prose; s_{xy} the sample covariance; s_D in Chapter 3's named-procedure table
Summaries	r, ρ	Sample and population correlation	Ch. 2	Partial correlation carries its conditioning set in a subscript (Ch. 2, Sec. 3.5; Ch. 8 develops multiple regression)
Relationships	r_{xyz}, r_s	Sample partial correlation controlling linearly for z ; the Spearman rank correlation	Ch. 2	Residual association, not causal exclusion; r_s is Pearson's coefficient on ranks
Population	μ, σ, σ^2	Process or population mean, SD, and variance; $\mu_1 - \mu_0$ a mean difference	Ch. 2	$\mu(x_0)$ the mean response at x_0 (Ch. 7)
Population	$p, \hat{p}, p_1, p_0$	A population proportion, its sample estimate, and group proportions	Ch. 3	Distinct from the p-value, written p-value in prose and $p = .0059$ for its value
Estimands	$\Delta\mu, \widehat{\Delta\mu}$	A difference in means as an estimand, $\Delta\mu = \mu_1 - \mu_0$ or the mean assignment effect over the stated participants, and its estimate $\widehat{\Delta\mu} = \bar{y}_1 - \bar{y}_0$; a stipulated value of $\Delta\mu$ serves planning and power calculations	Ch. 3	$\Delta p = p_1 - p_0$ for a difference in proportions (Ch. 6); θ , $\hat{\theta}$ for a generic estimand and estimator; τ and a bare Δ retired
Uncertainty	$SE(\cdot), \widehat{SE}(\cdot)$	The standard error of an estimator and its estimate; the hat marks the estimate. Name the statistic: $SE(\bar{y})$, $SE(\bar{y}_1 - \bar{y}_0)$, $SE(\bar{C})$	Ch. 3	Use the hat for an estimated SE, or explicitly state a locally abbreviated SE convention

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Uncertainty	$\text{Var}(\cdot)$, $\text{Cov}(\cdot, \cdot)$, $E(\cdot)$	Variance, covariance, and expectation; upright operator names with round brackets. Chapter 2 defines $E(X)$ for a probability model; Chapter 3 applies the same expectation to an estimator over stated repetitions, $E_{\text{assign}}(\cdot)$ when the repetition needs naming	Ch. 2; Ch. 3	$E[\cdot]$ and $\mathbb{E}$ retired
Probability	$P(\cdot)$, Φ	Probability of an event; the standard Normal distribution function	Ch. 2	ϕ its density
Distributions	m , λ , F_X , τ_K , μ_L , s_L	Binomial trial count; Poisson rate; CDF; Kendall correlation; logistic location and scale	Ch. 2	Use μ_L and s_L for the continuous logistic distribution; reserve $\ell(\cdot)$ for log-likelihood
Simulation	B	Number of simulation repetitions, distinct from a study's sample size n	Ch. 5	Monte Carlo SE shrinks with B ; the study SE does not; distinct from the bootstrap resample count B (Ch. 9, Sec. 6)
Reference distributions	z , $z_{1-\alpha/2}$	A standardized score and a standard Normal critical value	Ch. 2	—
Reference distributions	t_{df} , t^*	The t distribution with df degrees of freedom; a critical value	Ch. 3	t_{n-2} in regression
Reference distributions	χ^2 , $F(df_1, df_2)$	Chi-square and F reference distributions with their degrees of freedom	Ch. 3; Ch. 4	—
Testing	X^2	Observed Pearson chi-square statistic	Ch. 3	Distinguish it from the χ^2_{df} reference distribution
Reference distributions	df	Degrees of freedom, upright; subscripts only when two are needed	Ch. 2; Ch. 3	Fractional Welch degrees of freedom are reported to four decimals; Chapter 5 gives that quantity the symbol ν (row below)
Reference distributions	c^*	A generic critical value standing for $z_{1-\alpha/2}$ or t^* before the reference distribution is fixed	Ch. 5	Name the reference distribution at first use

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Reference distributions	ν	Welch degrees of freedom, $\nu = (a + b)^2 / [a^2 / (n_A - 1) + b^2 / (n_B - 1)]$ with $a = s_A^2 / n_A$ and $b = s_B^2 / n_B$, reported to four decimals	Ch. 5	A and B are the generic group labels, instantiated as A and M in Chapter 5's worked comparison; ordinary df elsewhere, including Chapter 3's fractional Welch value; Part One adopts ν for this quantity in Chapter 5
Testing	$H_0, H_1, \alpha, \beta, 1 - \beta$	Null and alternative hypotheses; Type I and Type II error probabilities; power at a specified alternative	Ch. 3	β here is distinct from a regression coefficient; H_A is the equivalent alternative-hypothesis label in Chapter 3
Magnitude	d, g	Standardized mean difference (Cohen's d , Hedges' g)	Ch. 3	—
ANOVA	$SS_B, SS_W, SS_T, MS_B, MS_W; SS_A, SS_{AB}, SS_E, MS_E$	Sums of squares and mean squares between, within, and total; factor, interaction and error terms of the two-way table; subscripted plain letters	Ch. 4	Not <code>\mathrm{}</code> words in subscripts; in the two-way table SS_B denotes factor B, not all between-cell variation (Ch. 4)
ANOVA	$\alpha_j, \beta_k, (\alpha\beta)_{jk}$	Factor-model main effects and interaction departures in the two-way model	Ch. 4	Distinct from the unindexed Type I and Type II error probabilities α and β
ANOVA	η^2, ω^2	ANOVA effect-size measures	Ch. 4	—
ANOVA	$c_j, C, \widehat{C}$	Contrast coefficients, a contrast $C = \sum_j c_j \mu_j$ with $\sum_j c_j = 0$, and its estimate $\widehat{C} = \sum_j c_j \bar{y}_j$	Ch. 4	L retired for contrasts; $L(\cdot)$ is a likelihood (Ch. 6)
ANOVA	$DD, \widehat{DD}$	A difference of differences between group means and its estimate	Ch. 6	Chapter 4 develops the object in words as a selected difference of differences; Chapter 10 writes the same object as C_{DD} when it is a linear combination of coefficients
Regression	$\beta_0, \beta_1, \beta_j; \widehat{\beta}_0, \widehat{\beta}_1$	Process intercept and slopes and their OLS estimates (<code>\widehat{}</code> , not <code>\hat{}</code>)	Ch. 7	Use <code>\widehat{\beta}</code> consistently

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Regression	$\hat{y}_i, e_i, \varepsilon_i$	Fitted value, residual, and error term of case i	Ch. 4; Ch. 7	<code>\widehat{y}</code> , not <code>\hat{y}</code> ; Chapter 4 uses the grouped forms $\hat{y}_{ij}, e_{ij}, \varepsilon_{ij}$ —
Regression	S_{xx}, S_{xy}	Centered sums of squares and cross-products	Ch. 7	—
Regression	$RSS, RegSS, TSS$	Residual, regression, and total sums of squares in regression; upright multi-letter names	Ch. 7	Distinct from the ANOVA SS family
Regression	s_e	Residual standard deviation (residual standard error)	Ch. 7	$\hat{\sigma}_{ML}$ is the maximum-likelihood scale, a different estimate (Ch. 8 Appendix B)
Regression	R^2	Coefficient of determination	Ch. 7	Adjusted R^2 named in words
Regression	$\hat{\mu}_0, Y_0$	The estimated mean response at x_0 and one new case there	Ch. 7	<code>\widehat{\mu}</code> , not <code>\hat{\mu}</code>
Regression	D, D_i, D_j	Indicator (0/1) variables; D_i marks case i 's membership, and $J - 1$ indicator columns D_j code a J -level factor with a named reference level	Ch. 2; Ch. 6; Ch. 8; Ch. 9	Chapter 2 writes the single group indicator D_i (Sec. 2.4); distinct from the deviance D (Ch. 11), Cook's distance D_i (Ch. 7–8), Chapter 3's random paired difference D_i , and the diagonal SD matrix D (Ch. 5)
Regression	$R\beta = r, q$	A set of q linear restrictions on the coefficients tested jointly	Ch. 9, Sec. 3	The restriction matrix is named at first use; distinct from the correlation matrix R and from R^2
Uncertainty	$B, \hat{\beta}_b^*$	Number of bootstrap resamples and the estimate from resample b	Ch. 9, Sec. 6	Recalled in Ch. 13; the same letter as the simulation repetition count of Ch. 5, declared locally at each use

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Matrices	$S, R, D; \widehat{V}$	Observed-variable covariance, correlation and diagonal SD matrices; estimated coefficient covariance is a different object	Ch. 5; Ch. 8	Define dimensions and the included records. For positive SDs, $R = D^{-1}SD^{-1}$ and $S = DRD$. Part One writes matrices as plain italic capitals, as the published Chapters 5, 7 and 8 do; Part Two's bold factor notation ($\mathbf{S}$, $\mathbf{\Lambda}$) is a separate convention.
Regression	$X, y, H, \hat{\beta}$	Design matrix, outcome vector, hat matrix, and coefficient vector; plain italic capitals, with $X'X$ for the cross-product	Ch. 7 App. A; Ch. 8 App. B; Ch. 9; Ch. 11	$h_{ii} = x_i'(X'X)^{-1}x_i$ as Chapters 7 and 8 write it
Checks	h_{ii}, D_i	Leverage and Cook's distance of case i	Ch. 7; Ch. 8	h_i in the one-predictor setting of Ch. 7
Checks	VIF_j	Variance inflation factor of predictor j	Ch. 8	—
Likelihood	$L(\cdot), \ell(\cdot), AIC, BIC$	Likelihood, log-likelihood, and information criteria	Ch. 6; Ch. 8 App. B; Ch. 9	Chapter 6 introduces a scalar likelihood; Chapter 8's Appendix B supplies the local Normal likelihood before AIC and BIC for its own criteria and homework; Chapter 9 develops the criteria; Chapter 11 develops likelihood fully
De-sign/contrasts	$\delta, \hat{\delta}$	A locally defined group contrast and its estimate	Ch. 8; Ch. 9	Conditional occupation-type contrast in Prestige; assignment contrast in STAR (Chapter 9). Causal interpretation requires design and assumptions.
Argument diagrams	E, J, I, C	Evidence, justification, inference and bounded claim: functions of information	Ch. 1; Sup. A	They name functions, not report headings or quality scores

Group	Symbol (preferred form)	Meaning	Main use	Alternatives and notes
Argument diagrams	STAT-E/J/I/C; SUB-E/J/I/C	The four functions in the statistical and substantive layers	Ch. 1; Sup. A/B	SUB-E situates, SUB-J augments and contextualizes, SUB-I interprets, SUB-C bounds the substantive reach of the corresponding statistical contribution
Argument diagrams	Support arrows and cross-layer links	Arrows show logical support within an argument; labelled links connect the meanings of the two layers	Ch. 1; Sup. A/B	Neither is a time sequence or causal path; label cross-question dependencies by the receiving function
Numbers	Displayed values	Calculated results to four decimals; the leading zero is kept except for bounded quantities (p , r , R^2 , proportions, probabilities: .0059); negative numbers in math mode for a true minus (−0.2179); no thousands separators; counts, ordinary df , and codes as integers	Ch. 1	Native software output keeps its own format
Prediction	MAE	Mean absolute error of held-out predictions	Ch. 8	—